ABRA: Agent-Bound UI for Human–Agent Communication beyond Chat with Reusable Views

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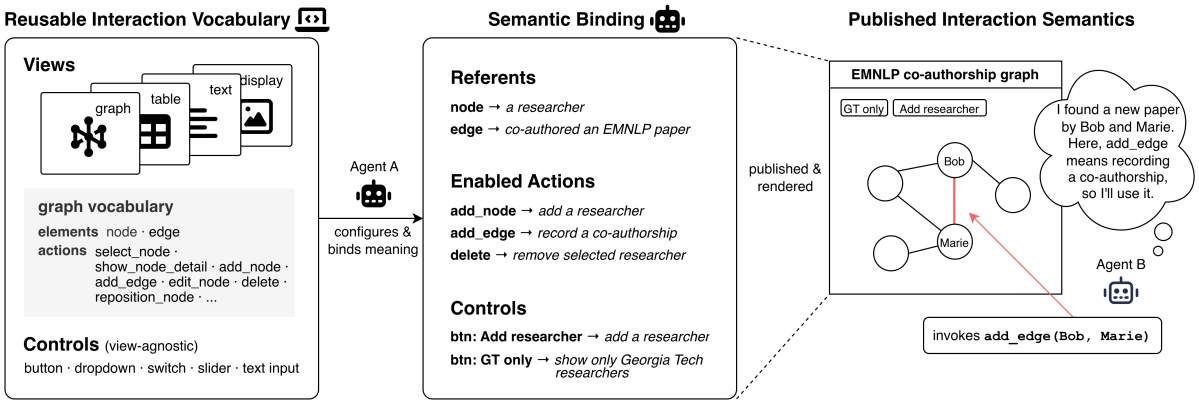


Figure 1: Agent-Bound UI separates a reusable interaction vocabulary (§3.1) from task-specific semantic bindings (§3.2). The agent publishes those bindings so that humans and other agents can read and act on them (§3.3).

Abstract

Agents can create rich representations, yet these representations are rarely used as a medium for expressing user intent. We present *Agent-Bound UI*, an interaction model that separates reusable interaction structures from task-specific meanings. Agents bind task-specific meanings to interactions in reusable views and publish the resulting bindings as shared state, enabling humans and agents to understand and act on shared surfaces whose meanings vary across tasks. We implement ABRA, a framework-agnostic prototype of Agent-Bound UI, in which users and agents collaborate through chat and shared interactive canvases. We demonstrate ABRA through a representative use case and evaluate four LLM-backed agents on canvas authoring, owner-side consequence resolution, and cross-agent invocation. The code and demo video are available at https://github.com/megagonlabs/abra.

1 Introduction

Agentic AI is increasingly used to assist users with a wide range of tasks (Wang et al., 2024b), from everyday activities such as drafting emails and planning trips to professional work including cleaning datasets (Chen et al., 2026b), curating knowledge graphs (Pan et al., 2024), and writing code (Liu et al., 2026). Unlike conventional conversational assistants that primarily answer questions, modern agents are connected to external tools and services to actively carry out these tasks, often across multiple coordinating agents. In this setting, chat serves as the primary medium through which humans and agents collaborate. Users specify task goals, review and revise plans, inspect intermediate results, and provide feedback throughout the task lifecycle.

Meanwhile, the way agents present their work to users has steadily evolved. Early agents responded in plain text (OpenAI et al., 2024), markdown has since become the standard (Chen et al., 2025), and more recently interactive HTML-based representations (e.g., artifacts, canvases, and inline widgets) have begun gaining traction (Shihipar, 2026). For example, an agent may present a plan as a flowchart or a set of results as a table rather than as text.

Yet while representations from agents to humans have become increasingly rich, the human-to-agent channel remains largely unchanged, with chat continuing to serve as the dominant interface for user input. This creates a *representation–interaction asymmetry*: the output side is increasingly structured and multimodal, whereas the input side remains constrained to natural language.

This asymmetry, however, reveals an opportunity: rich representations are already rendered on the user's screen. We therefore ask whether these